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Course code:DSA0602

Class Activity 2: 15.07.2026

Scenario 1:Smart City Air Quality Monitoring.

Python Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Read the dataset
data = pd.read_csv("smart_city_air_quality.csv")

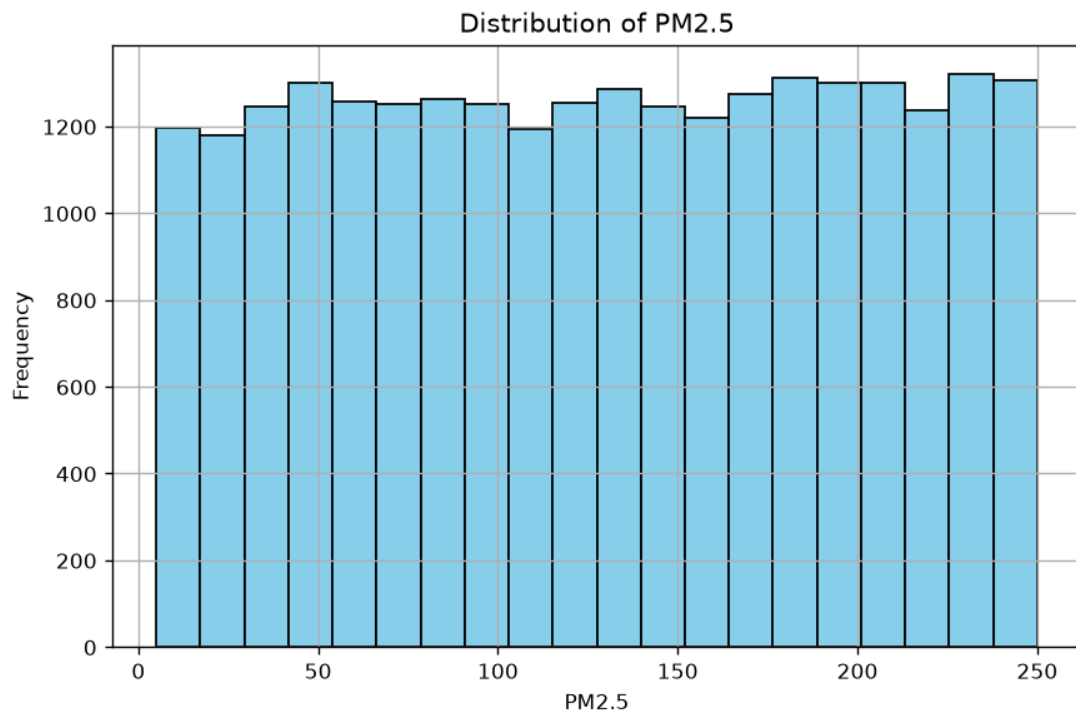
# Plot Histogram
plt.figure(figsize=(8,5))
plt.hist(data["PM2_5"], bins=20, color="skyblue", edgecolor="black")

plt.title("Distribution of PM2.5")
plt.xlabel("PM2.5")
plt.ylabel("Frequency")

plt.grid(True)

# Show the graph
plt.show()
```

Output:



Scenario 2: Earthquake Risk Analysis

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt

# Read the dataset

data = pd.read_csv("earthquake.csv")

# Create Bubble Chart

plt.figure(figsize=(10,6))

plt.scatter(

    data["Longitude"],

    data["Latitude"],

    s=data["Magnitude"]*50,

    c=data["Depth"],

    cmap="viridis",

    alpha=0.7,

    edgecolors="black"

)

# Color bar

plt.colorbar(label="Depth (km)")
```

```
# Titles and Labels
```

```
plt.title("Earthquake Risk Analysis")
```

```
plt.xlabel("Longitude")
```

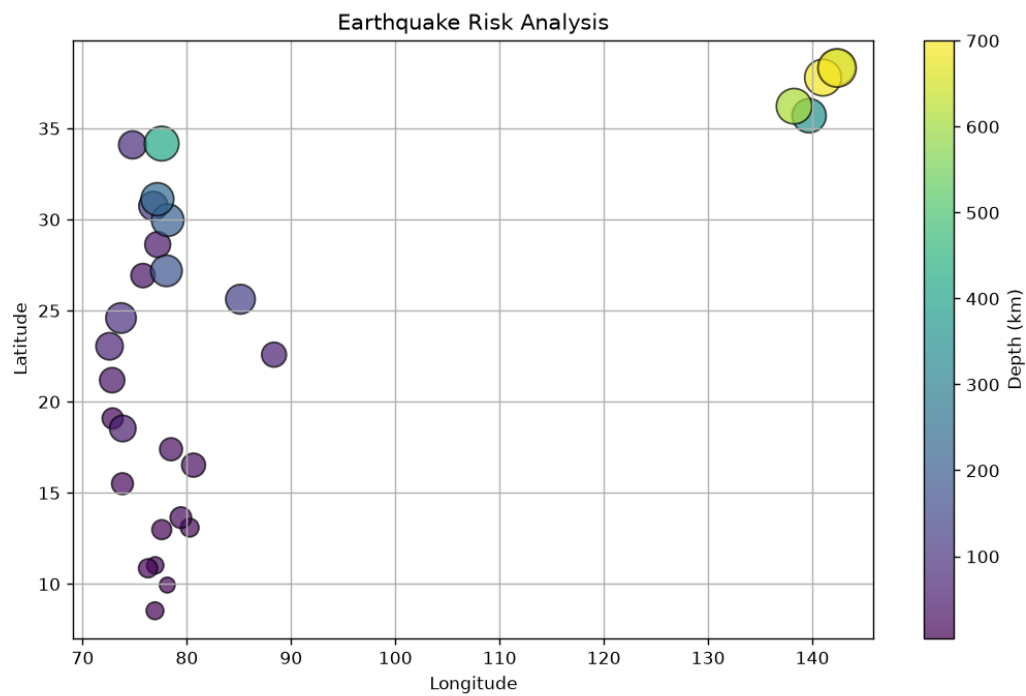
```
plt.ylabel("Latitude")
```

```
plt.grid(True)
```

```
# Display Graph
```

```
plt.show()
```

Output:



Scenario 3: Wind Direction Analysis for Airport Operations

Python code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Read dataset
df = pd.read_csv("airport_wind.csv")

# Overall Wind Direction
plt.figure(figsize=(8,5))
plt.scatter(df["Wind_Direction"], df["Wind_Speed_kmh"])
plt.title("Airport Wind Direction Analysis")
plt.xlabel("Wind Direction (Degrees)")
plt.ylabel("Wind Speed (km/h)")
plt.grid(True)
plt.show()

# Seasonal (Monthly) Variation
months = df["Month"].unique()

plt.figure(figsize=(10,6))

for month in months:
    data = df[df["Month"] == month]
```

```
plt.scatter(data["Wind_Direction"], data["Wind_Speed_kmh"], label=month)
```

```
plt.title("Wind Direction by Month")
```

```
plt.xlabel("Wind Direction (Degrees)")
```

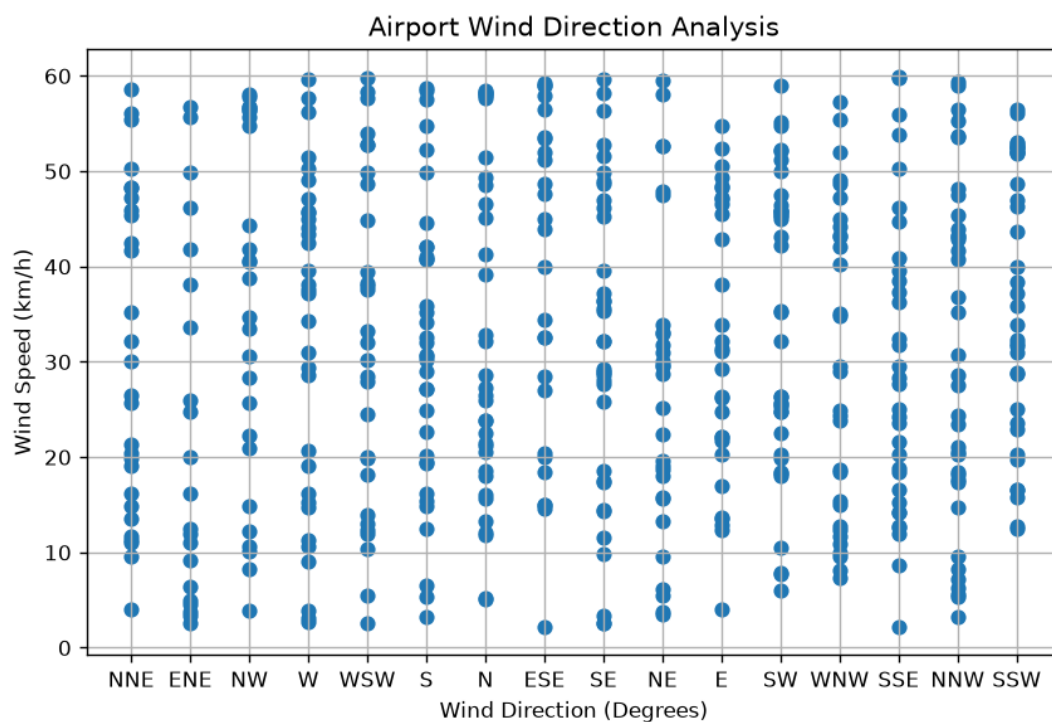
```
plt.ylabel("Wind Speed (km/h)")
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.show()
```

Output:



Scenario 4: Hospital Patient Monitoring Dashboard

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt

# Read dataset

df = pd.read_csv("hospital_patients.csv")

plt.figure(figsize=(12,8))

# Heart Rate

plt.subplot(3,2,1)

plt.plot(df["Heart_Rate_bpm"])

plt.title("Heart Rate")

# Oxygen Saturation

plt.subplot(3,2,2)

plt.plot(df["Oxygen_Saturation_%"])

plt.title("Oxygen Saturation")

# Systolic Blood Pressure

plt.subplot(3,2,3)

plt.plot(df["Systolic_BP"])

plt.title("Systolic BP")
```

```
# Diastolic Blood Pressure
```

```
plt.subplot(3,2,4)
```

```
plt.plot(df["Diastolic_BP"])
```

```
plt.title("Diastolic BP")
```

```
# Respiration Rate
```

```
plt.subplot(3,2,5)
```

```
plt.plot(df["Respiration_Rate"])
```

```
plt.title("Respiration Rate")
```

```
# Temperature
```

```
plt.subplot(3,2,6)
```

```
plt.plot(df["Temperature_C"])
```

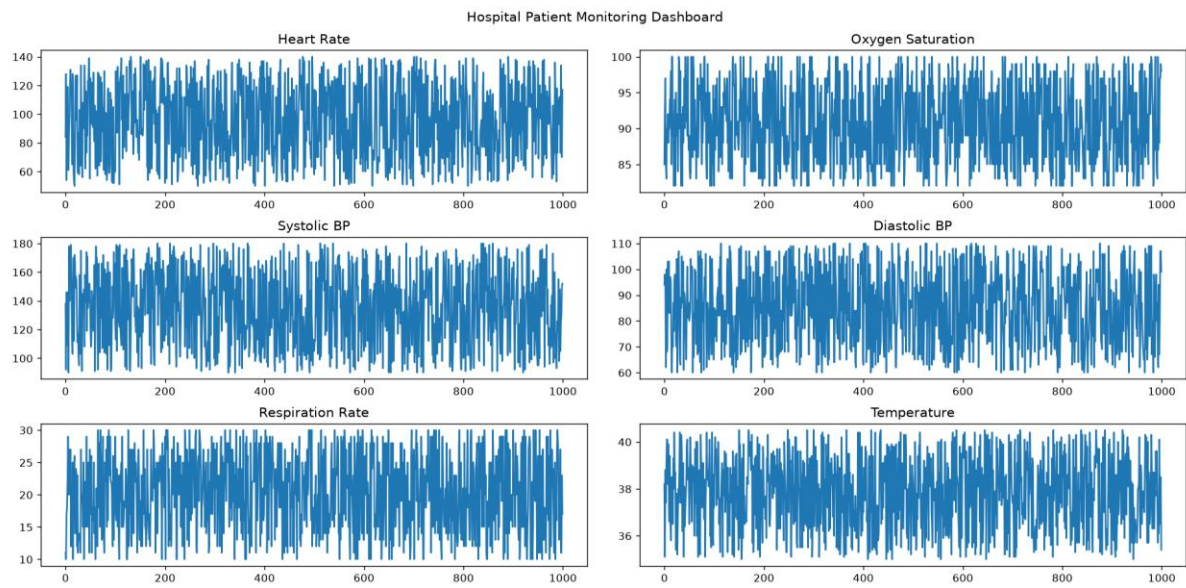
```
plt.title("Temperature")
```

```
plt.suptitle("Hospital Patient Monitoring Dashboard")
```

```
plt.tight_layout()
```

```
plt.show()
```

Output:



Scenario 5: Global Climate Change Study

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt

# Read CSV

df = pd.read_csv("global_climate_change.csv")

# Check first 5 rows

print(df.head())

countries = df["Country"]
anomaly = df["Temperature_Anomaly_C"]

# Colors

colors = ["red" if x > 0 else "blue" for x in anomaly]

# Plot

plt.figure(figsize=(10,6))

plt.barh(countries, anomaly, color=colors)

plt.axvline(0, color="black")

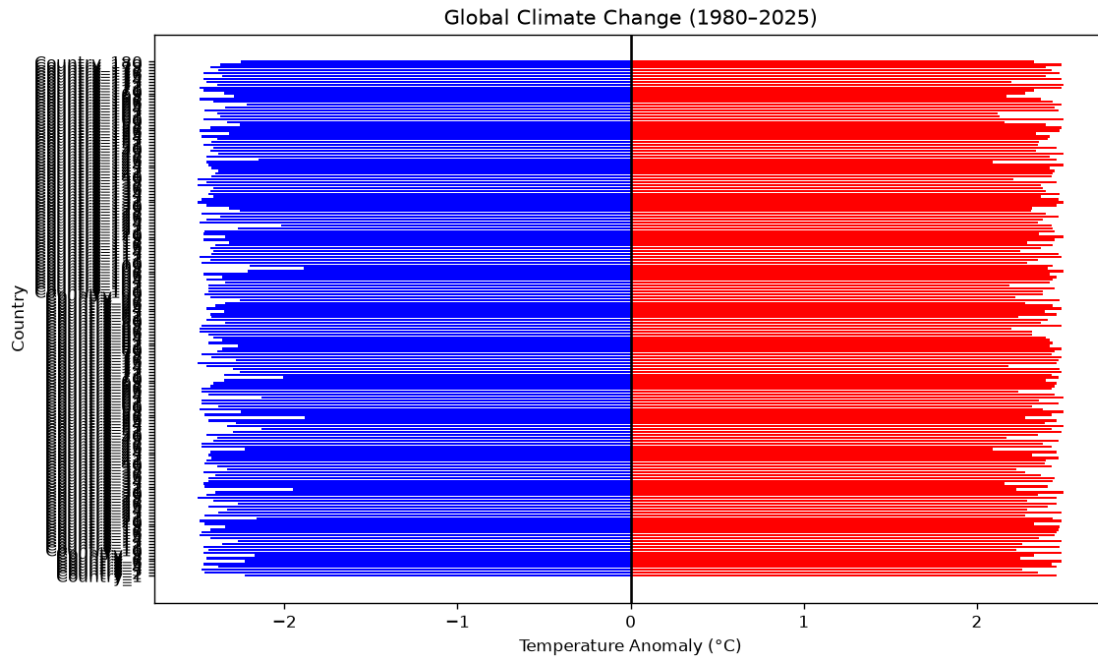
plt.title("Global Climate Change (1980–2025)")

plt.xlabel("Temperature Anomaly (°C)")
```

```
plt.ylabel("Country")
```

```
plt.show()
```

Output:



Scenario 6: E-Commerce Sales Analytics

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt


# Read dataset

df = pd.read_csv("ecommerce_sales.csv")


# Sum sales by category

sales = df.groupby("Category")["Total_Sales"].sum()


# Plot

plt.figure(figsize=(8,5))

plt.bar(sales.index, sales.values)


plt.title("Total Sales by Category")

plt.xlabel("Category")

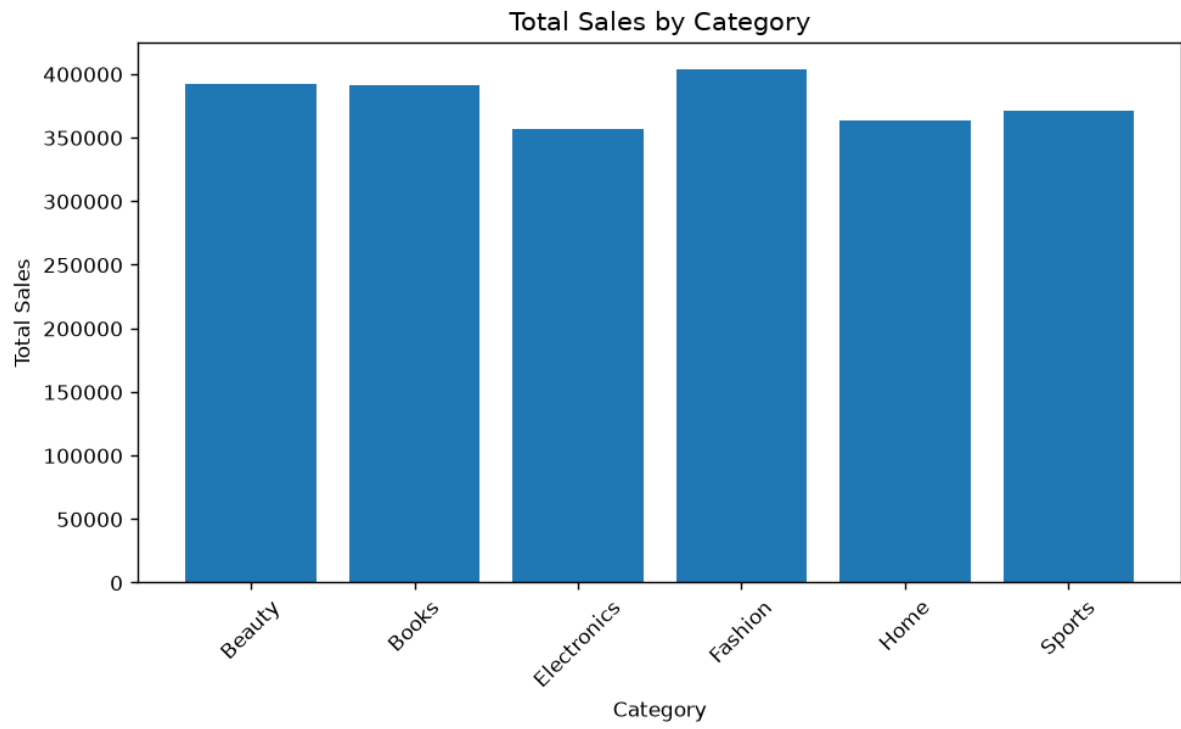
plt.ylabel("Total Sales")

plt.xticks(rotation=45)


plt.tight_layout()

plt.show()
```

Output:



Scenario 7: Satellite Image Vegetation Analysis

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt

# Read dataset

df = pd.read_csv("satellite_ndvi.csv")

# Classify NDVI values

colors = []

for x in df["NDVI"]:

    if x > 0.5:

        colors.append("green")

    elif x >= 0:

        colors.append("yellow")

    else:

        colors.append("blue")

# Plot

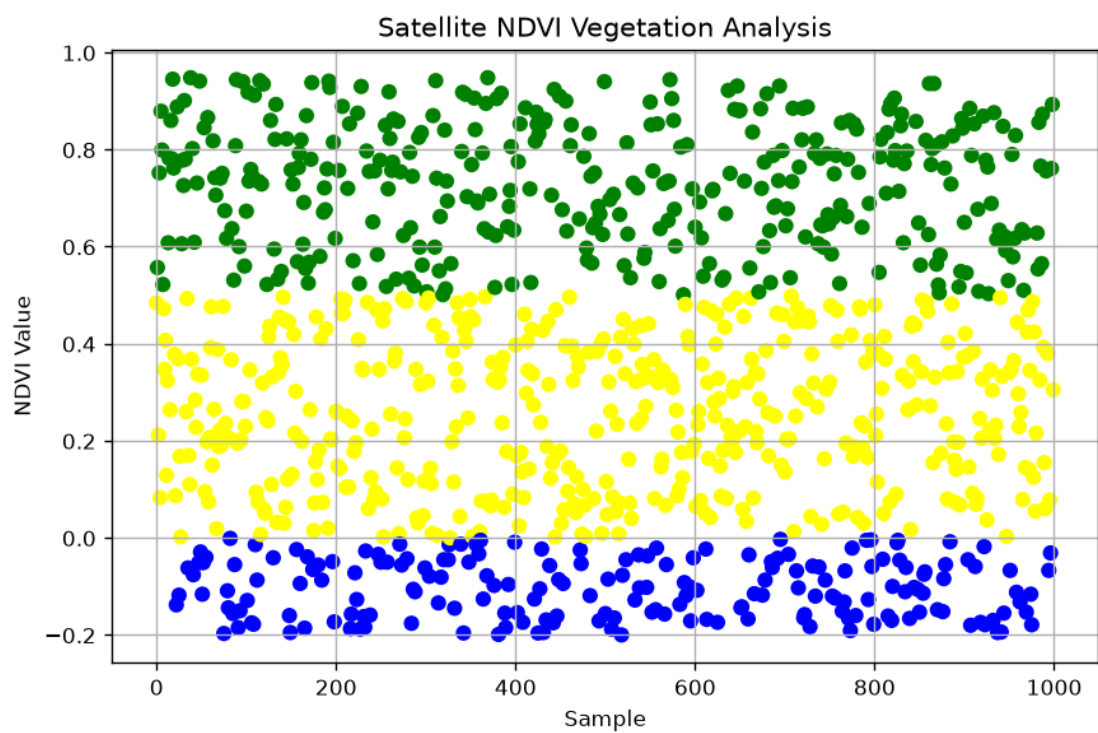
plt.figure(figsize=(8,5))

plt.scatter(range(len(df)), df["NDVI"], c=colors)

plt.title("Satellite NDVI Vegetation Analysis")
```

```
plt.xlabel("Sample")  
plt.ylabel("NDVI Value")  
plt.grid(True)  
  
plt.show()
```

Output:



Scenario 8: Cryptocurrency Market Volatility

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt

# Read dataset

df = pd.read_csv("cryptocurrency_market.csv")

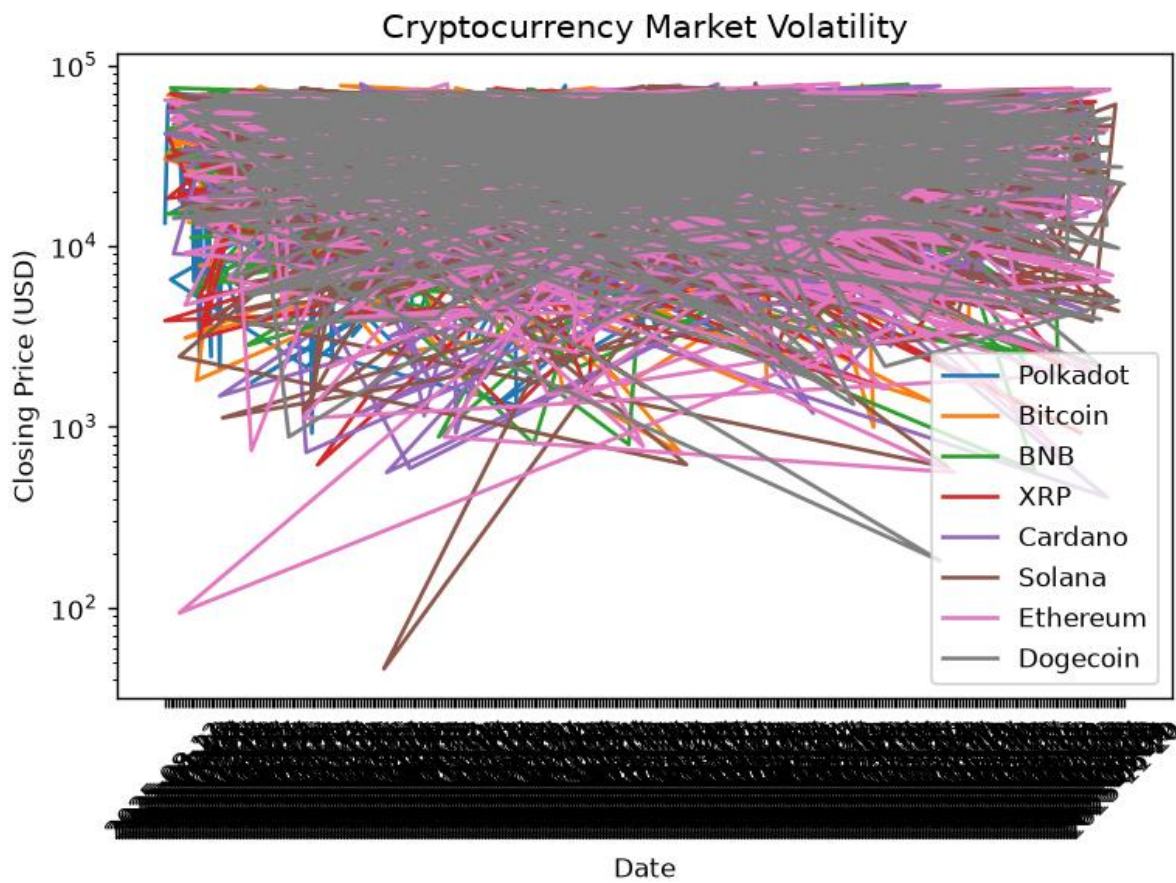
# Plot each cryptocurrency
for crypto in df["Cryptocurrency"].unique():
    data = df[df["Cryptocurrency"] == crypto]
    plt.plot(data["Date"], data["Close_Price_USD"], label=crypto)

# Logarithmic scale
plt.yscale("log")

plt.title("Cryptocurrency Market Volatility")
plt.xlabel("Date")
plt.ylabel("Closing Price (USD)")
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()

plt.show()
```

Output:



Scenario 9: Marathon Runner Performance Analysis

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt

# Read dataset

df = pd.read_csv("marathon_runner.csv")

# Scatter Plot

plt.figure(figsize=(8,5))

plt.scatter(

    df["Distance_km"],

    df["Average_Pace_min_per_km"],

    c=df["Average_Heart_Rate_bpm"],

    s=60

)

plt.colorbar(label="Average Heart Rate (BPM)")

plt.title("Marathon Runner Performance Analysis")

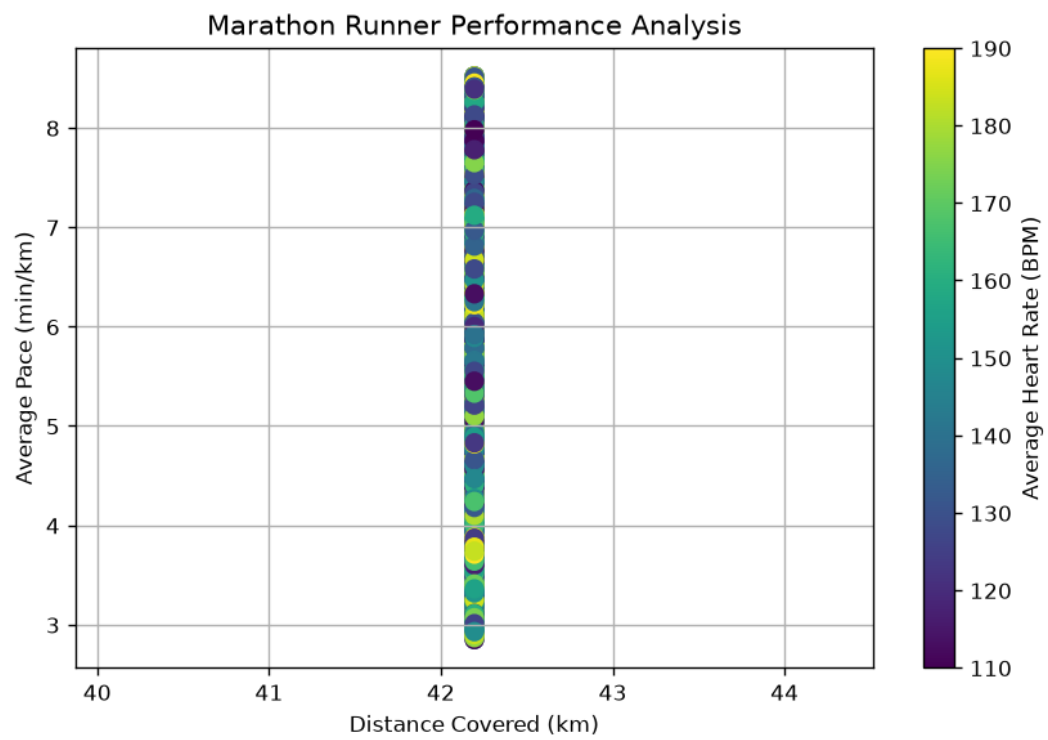
plt.xlabel("Distance Covered (km)")

plt.ylabel("Average Pace (min/km)")

plt.grid(True)
```

```
plt.show()
```

Output:



Scenario 10: COVID-19 Vaccination Campaign Dashboard

Python code:

```
import pandas as pd

import matplotlib.pyplot as plt

# Read CSV

df = pd.read_csv("covid_vaccination.csv")

# Convert Date column

df["Date"] = pd.to_datetime(df["Date"])

# Group by Date

daily = df.groupby("Date")[["People_Fully_Vaccinated",
                             "People_Partially_Vaccinated"]].sum()

# Plot

plt.figure(figsize=(12,6))

plt.plot(daily.index,
         daily["People_Fully_Vaccinated"],
         marker='o',
         linewidth=2,
         label="Fully Vaccinated")
```

```
plt.plot(daily.index,  
         daily["People_Partially_Vaccinated"],  
         marker='s',  
         linewidth=2,  
         label="Partially Vaccinated")  
  
plt.title("COVID-19 Vaccination Campaign Dashboard")  
plt.xlabel("Date")  
plt.ylabel("People Vaccinated")  
plt.legend()  
plt.grid(True)  
  
plt.xticks(rotation=45)  
plt.tight_layout()  
  
plt.show()
```

Output:

